EXERCÍCIOS

I.Derive as funções

(a) $f(t) = t^8 - 2t^5 + 3t + 1$

(b) $h(x) = \left(\frac{1}{x^2} + 3\right).\left(\frac{2}{x^3} + x\right)$

(c) $y = (2x^2 + x - 5)^3$

d) $h(u) = \sqrt{\frac{2u+1}{u-1}}$

(e) $p(u) = \frac{2}{3}(5u-3)^{-1}(5u+3)$

(f) $g(t) = \frac{1}{3t^3} - \frac{1}{2t^2} + 1$

(g) $p(x) = \frac{2x^2 + x + 1}{x^2 - 3x + 2}$

(h) $h(t) = \sqrt{2t^3 - t + 1}$

(i) $g(s) = (4s^2 - 5s + 2)^{-1/3}$

(j) $f(y) = (5y - 2)^6 (3y - 1)^3$

(k) $h(y) = \frac{(y-1)^3}{(y-2)^2(y-3)}$

II. Derive as funções

(a) $f(x)= \frac{sen(5x)}{\cos(x)}$

(b) $p(u) = u^5 \, tgu$

(c) $f(x) = \left(\frac{1}{sen\,x}\right)^2$

(d) $g(x) = 3x\,(2tgx+1) + \sqrt{x}$

(e) $g(t) = 2\cos(2t^2 - 3t + 1)$

(f) **y =** e^{x^2+3x}

(g) $h(t) = \dfrac{\sqrt{t} + \operatorname{sen} t}{\operatorname{sen} t}$

(h) $g(v) = \dfrac{\operatorname{sen}(v^3 - 2)}{\operatorname{sen} 2v}$

(i) $(y) = \dfrac{y4 + y}{\cos 2y}$

(j) $f(u) = \operatorname{sen}^2\left(\dfrac{u}{2}\right).\cos^2\left(\dfrac{u}{2}\right)$

(k) y = ln(t2+4)

(l) y = 22x - 1